TLP:CLEAR

GOOGLE GEMMA 4

Technical Summary

Architecture, Licensing, Deployment & Benchmarks

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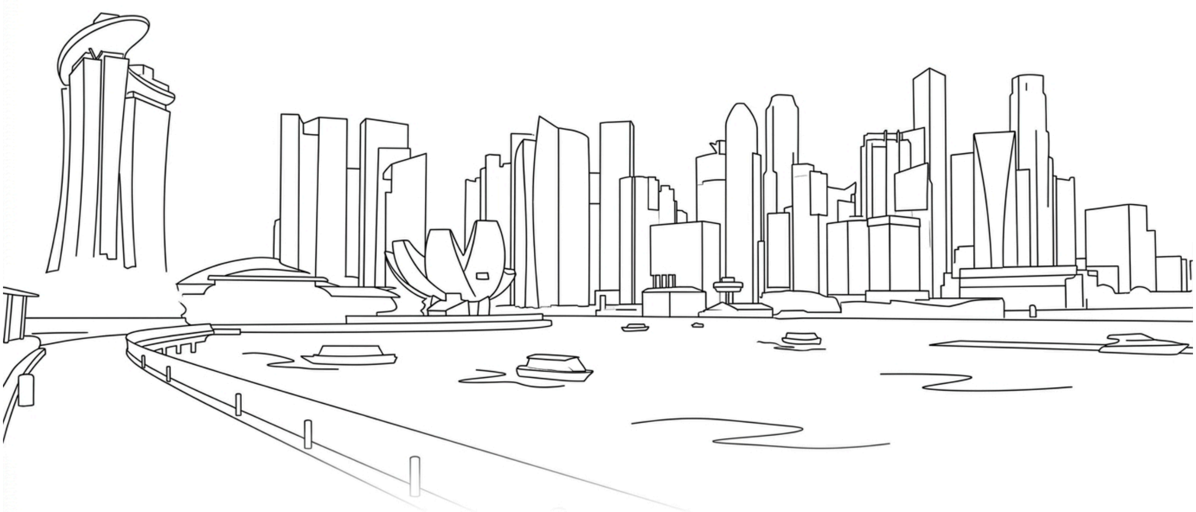

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Executive Summary

On 2 April 2026, Google DeepMind released Gemma 4, a family of four open-weight multimodal models derived from the same research underpinning the Gemini 3 model line. Gemma 4 is the first release in the Gemma series to adopt the Apache License 2.0, replacing the previous custom Gemma license that imposed usage restrictions and permitted unilateral term modifications by Google.

The family spans four model sizes — from sub-3B edge models to a 31B dense model — supporting text, image, and audio modalities with context windows up to 256K tokens. All variants are available via Hugging Face, Kaggle, and Ollama with day-one support across major inference frameworks, including llama.cpp, vLLM, MLX, and Ollama.

1. Licensing Change: Apache 2.0

Prior Gemma releases (Gemma 1 through Gemma 3) shipped under a proprietary “Gemma Terms of Use” license that restricted certain commercial use cases, included “Harmful Use” carve-outs that required legal interpretation, and allowed Google to modify the terms post-release. Notably, the Gemma source code (e.g., gemma.cpp) was already under Apache 2.0, but the **model weights** — the commercially relevant artefact — were governed by the restrictive custom license.

Gemma 4 moves both the weights and the code under the OSI-approved Apache License 2.0. Multiple independent sources confirm this applies without residual restrictions: no monthly active user caps, no acceptable-use policy enforcement clauses, no “Restricted Topics” carve-outs, and no restrictions on redistribution or commercial deployment. The LICENSE file in the google-deepmind/gemma repository contains the standard Apache 2.0 text with no addenda. This aligns Gemma’s licensing posture with competing open-weight model families, including Alibaba’s Qwen, Mistral, and Arcee, removing a significant friction point for enterprise legal review.

2. Model Family Architecture

Gemma 4 comprises four models organised into two deployment tiers. All models share a 262K-token vocabulary. Specifications below are from the official Google Gemma 4 model card.

2.1 Dense Models

Property	E2B	E4B	31B Dense
Total Params	5.1B (2.3B eff.)	8B (4.5B eff.)	30.7B
Layers	35	42	60
Sliding Window	512 tokens	512 tokens	1024 tokens
Context Length	128K tokens	128K tokens	256K tokens
Modalities	Text, Image, Audio	Text, Image, Audio	Text, Image
Vision Encoder	~150M params	~150M params	~550M params
Audio Encoder	~300M params	~300M params	None

2.2 Mixture-of-Experts (MoE) Model

Total Parameters	25.2B
Active Parameters	3.8B
Layers	30
Sliding Window	1024 tokens
Context Length	256K tokens

Expert Config	128 total / 8 active + 1 shared per token
Modalities	Text, Image
Vision Encoder	~550M params

Each layer runs a dense GeGLU FFN in parallel with the 128-expert MoE block, and the outputs are summed and scaled by $1/\sqrt{2}$. Selective activation enables compute throughput comparable to a 4B model; however, all 25.2B parameters must remain resident in memory.

2.3 Proportional RoPE (p-RoPE) — The Mathematics of Long Context

The Problem: Mechanical Amnesia at Scale. Standard RoPE encodes position by rotating embedding vectors at different frequencies across dimensions. High-frequency dimensions encode precise local position; low-frequency dimensions encode broad semantic relationships. At 256K context, even the slowest-rotating dimensions accumulate enough angular displacement to wrap around and point backwards. The dot product between semantically related tokens separated by hundreds of thousands of positions becomes negative — the model perceives related concepts as mathematical opposites. This is described in Theorem 6.1 as a “geometric trap” in which long-distance semantic alignment fails via phase inversion rather than gradual degradation.

The Solution: Freezing the Semantic Channels. Proportional RoPE fixes this by truncating the slowest rotations to zero. In Gemma 4’s global attention layers, only 25% of embedding dimensions receive rotary encoding (partial=0.25). The remaining 75% are frozen at zero rotation — they never rotate regardless of token distance.

Channel	Dimensions	Rotation	Function
Positional	25% of head_dim	RoPE (θ=1M)	“Where is this token?” — positional awareness
Semantic	75% of head_dim	Frozen (zero)	“What is this token about?” — distance-agnostic meaning

The semantic channel dimensions produce identical dot products whether tokens are 10 or 250,000 positions apart. They are pure content-matching channels that never suffer phase degradation.

Why p-RoPE only applies to global layers: Sliding-window layers (5 of every 6) attend within 512–1024 tokens, where standard RoPE (θ=10K) works perfectly. Global layers (1 of every 6) attend across the full 256K context and require p-RoPE to avoid the geometric trap. The final layer is always global, ensuring full visibility of context at the model’s last computation.

2.4 Additional Architectural Features

Unified Keys and Values (K=V Weight Sharing). In global attention layers, the V projection is eliminated; unified key-value tensors reduce both compute and memory for long-context generation.

Shared KV Cache. The last N layers (num_kv_shared_layers) reuse K and V tensors from the last non-shared layer of the same attention type. In E2B, 20 of 35 layers share KV states.

Per-Layer Embeddings (PLE). Edge models incorporate a second embedding table mapping tokens to (layers × 256) dimensions. Each decoder layer receives a unique 256-dim slice, gated and added as a residual.

3. Capabilities

Configurable Thinking Mode. All Gemma 4 models include a built-in reasoning mode activated by including the <|think|> token at the start of the system prompt. When enabled, the model outputs internal reasoning within <|channel>thought tags before the final answer. Google has not disclosed whether this is backed by RL (e.g., GRPO) or by distillation from Gemini 3.

Agentic Workflows. Native function calling with structured JSON output and native system prompt support (standard system/assistant/user roles) enables building autonomous agents without grammar-constrained generation.

Multimodal Input. All models process text and images with variable aspect ratios and configurable visual token budgets (70, 140, 280, 560, 1120 tokens per image). Edge-tier models additionally support native audio input (up to 30 seconds) via a USM-style conformer encoder.

Additional Capabilities. Code generation and correction, interleaved multimodal input, over 140 languages natively, and GUI element detection with bounding box output in JSON format.

4. Local Inference Quick-Start Guide

4.1 Hardware Requirements

Critical caveat: The figures below represent model weights only and do not include the KV cache, which scales linearly with context length. At 256K context on the 31B model, the KV cache alone can consume 10–15 GB or more. A consumer GPU with 24 GB VRAM running the 31B at Q4 will likely OOM well before reaching the full 256K context window.

Model	4-bit (Q4_K_M)	8-bit (Q8_0)	Full Precision (BF16)
E2B	~3 GB	~5 GB	~15 GB
E4B	~4 GB	~5 GB	~15 GB
26B A4B (MoE)	~18 GB	~28 GB	Single 80 GB H100
31B Dense	~20 GB	~34 GB	Single 80 GB H100

4.2 Quick-Start: llama.cpp

llama.cpp requires build b8636 or later for full support for Gemma 4.

Step 1 — Install or update llama.cpp

macOS (Homebrew):

```
brew upgrade llama.cpp
# If latest build unavailable:
brew install llama.cpp --HEAD
```

Linux / Build from source:

```
git clone https://github.com/ggml-org/llama.cpp
cmake llama.cpp -B llama.cpp/build -DBUILD_SHARED_LIBS=OFF -DGGML_CUDA=ON
cmake --build llama.cpp/build --config Release -j --clean-first
```

For Apple Silicon (Metal), set -DGGML_CUDA=OFF; Metal acceleration is enabled by default. For AMD GPUs, use -DGGML_HIP=ON.

Step 2 — Select and run by hardware profile

Hardware	Recommended Command
MacBook 16 GB	`llama-server -hf ggml-org/gemma-4-E4B-it-GGUF:Q8_0`
MacBook 24 GB+	`llama-server -hf ggml-org/gemma-4-26B-A4B-it-GGUF:Q4_K_M`
RTX 3090 (24 GB)	`llama-server -hf ggml-org/gemma-4-26B-A4B-it-GGUF:Q4_K_M`
RTX 5090 (32 GB)	`llama-server -hf ggml-org/gemma-4-26B-A4B-it-GGUF:Q8_0`
H100 (80 GB)	`llama-server -hf ggml-org/gemma-4-31B-it-GGUF:F16`

The llama-server command automatically downloads the GGUF checkpoint and starts an OpenAI-compatible API endpoint at http://localhost:8080/v1.

4.3 Quick-Start: Ollama

Ollama provides the fastest path to running Gemma 4 locally with minimal configuration. Download from https://ollama.com.

```
ollama run gemma4          # Default (auto-selects best variant)
ollama run gemma4:e4b      # Edge — laptops with 8+ GB RAM
ollama run gemma4:26b      # MoE — consumer GPUs with 24+ GB
ollama run gemma4:31b      # Dense — workstations with 32+ GB
```

4.4 Quick-Start: MLX (Apple Silicon)

```
curl -fsSL https://raw.githubusercontent.com/unslothai/unsloth/
  refs/heads/main/install_gemma4_mlx.sh | sh
source ~/.unsloth/unsloth_gemma4_mlx/bin/activate
python -m mlx_lm chat --model unsloth/gemma-4-E4B-it-UD-MLX-4bit
```

4.5 Quick-Start: Python (Hugging Face Transformers)

```
from transformers import pipeline
pipe = pipeline("any-to-any", model="google/gemma-4-e4b-it")
result = pipe({"text": "Explain quantum computing."})
```

4.6 Recommended Inference Parameters

Google's default sampling configuration: temperature 1.0, top-p 0.95, top-k 64. Repetition and presence penalties should remain at 1.0 unless token repetition is observed. To enable chain-of-thought reasoning, include `<|think|>` at the start of the system prompt.

4.7 Connecting to Agent Frameworks

Both llama-server and Ollama expose OpenAI-compatible API endpoints, enabling direct integration with agent frameworks without custom adapters. Verified compatible frameworks include OpenClaw, Hermes, Pi, and Open Code.

4.8 Additional Supported Runtimes

Day-one support: vLLM, mistral.rs (Rust-native), SGLang, NVIDIA NIM, LM Studio, Unsloth Studio, Keras, Baseten, Docker, MaxText, and transformers.js (browser inference via WebGPU). AMD GPUs supported via ROCm through vLLM, SGLang, llama.cpp, and Lemonade Server.

5. Benchmark Results (Official)

All results below are for instruction-tuned models, sourced from the official Gemma 4 model card.

5.1 Text Benchmarks

Benchmark	31B	26B MoE	E4B	E2B	Gemma 3 27B
MMLU Pro	85.2%	82.6%	69.4%	60.0%	67.6%
AIME 2026	89.2%	88.3%	42.5%	37.5%	20.8%
LiveCodeBench v6	80.0%	77.1%	52.0%	44.0%	29.1%
Codeforces Elo	2150	1718	940	633	110
GPQA Diamond	84.3%	82.3%	58.6%	43.4%	42.4%
BigBench XH	74.4%	64.8%	33.1%	21.9%	19.3%
MMMLU	88.4%	86.3%	76.6%	67.4%	70.7%

5.2 Vision Benchmarks

Benchmark	31B	26B MoE	E4B	E2B	Gemma 3 27B
MMMU Pro	76.9%	73.8%	52.6%	44.2%	49.7%
MATH-Vision	85.6%	82.4%	59.5%	52.4%	46.0%
MedXPertQA MM	61.3%	58.1%	28.7%	23.5%	—

The 31B's BigBench Extra Hard score of 74.4% represents a nearly 4× improvement over Gemma 3 27B's 19.3%. On the Arena AI text leaderboard, the 31B ranked #3 (Elo 1452) and the 26B MoE ranked #6 (Elo 1441).

6. Prompt Speculative Decoding (300 t/s Demo)

On 2 April 2026, Georgi Gerganov (creator of llama.cpp) demonstrated Gemma 4 26B A4B Q8_0 running on a Mac Studio M2 Ultra (3-year-old hardware) with llama.cpp's built-in WebUI, MCP support (web-search, HuggingFace, GitHub tools), and **prompt speculative decoding** — achieving 300 tokens/second in a real-time video.

How it works: This is distinct from traditional two-model speculative decoding. When llama.cpp's MCP integration injects tool results (thousands of tokens of search results) into the prompt, and the model then generates a response referencing that content, llama.cpp proposes candidate token sequences drawn from the prompt itself as "draft" tokens, then batch-verifies them in a single forward pass. Since MCP tool results are often quoted or paraphrased, the acceptance rate is extremely high.

Critical caveat: The 300 t/s represents peak burst throughput during prompt-recitation phases. For purely novel token generation, throughput drops to approximately 30–60 t/s on that hardware. Gerganov posted a follow-up caveat acknowledging the prompt-recitation component.

Significance for agentic workflows: This is particularly relevant for RAG and CTI workflows where the model ingests retrieved content via MCP tools and then produces structured reports — the inherent "retrieve → summarise" pattern maximises prompt speculative decoding acceptance rates.

7. Competitive Context

Gemma 4 enters a competitive open-weight landscape. Over the preceding year, Chinese open-source model families — notably Alibaba's Qwen, DeepSeek, and GLM — gained significant market share, with Qwen overtaking Meta's Llama as the most-used self-hosted model globally by late 2025.

Model	Arena Elo	GPQA Diamond	MMLU Pro	Active Params
Gemma 4 31B Dense	1452 (#3)	85.7%	85.2%	31B
Gemma 4 26B MoE	1441 (#6)	79.2%	—	3.8B
Qwen 3.5 27B	—	85.8%	—	27B
Llama 4 Scout	—	—	—	17B (10M ctx)

Google has indicated that additional Gemma 4 model sizes may follow the initial release. The E2B and E4B models serve as the foundation for Gemini Nano 4, Google's next-generation on-device model for Android, expected on consumer devices later in 2026.

8. Open Technical Questions

Training methodology for reasoning. Google has not disclosed whether the thinking/reasoning capability uses RL (GRPO/PPO), distillation from Gemini 3, or another approach. Preserving reasoning capability during SFT requires mixing reasoning-style and direct-answer examples (Unsloth recommends a minimum of 75% reasoning-style examples).

Speculative decoding. No official draft-model pairing has been announced (e.g., using E2B as a drafter for the 31B). Community experimentation is expected.

Tokeniser compatibility. Google has not confirmed whether Gemma 4 uses the same tokeniser as Gemma 3 (the Gemini 2.0 tokeniser with a 256K vocabulary). If changed, existing vector database embeddings would require full re-indexing for RAG pipelines.

Full technical report. As of 3 April 2026, no Gemma 4 technical report has been published. The Gemma 3 Technical Report (arXiv:2503.19786) remains the most recent. Google I/O 2026 (expected in May) may provide additional details.

9. Model Weights & Download Links

Source	URL
Hugging Face (Official)	https://huggingface.co/collections/google/gemma-4
Hugging Face (GGUF — Unsloth)	https://huggingface.co/collections/unsloth/gemma-4-gguf
Kaggle	https://www.kaggle.com/models/google/gemma-4
Ollama	https://ollama.com/library/gemma4
Google AI Studio (31B, 26B)	https://aistudio.google.dev
Google AI Edge Gallery (E4B, E2B)	Via Android app

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